

A DOE STUDY ON THE HYPERGOLICITY OF HYDROGEN PEROXIDE WITH A ROCKET LIQUID FUEL BASED ON MONOETHANOLAMINE AND ETHANOL

Leandro José Maschio,¹ Luís Gustavo Ferroni Pereira,² William Müller Meyer,¹ Rodrigo Intini Marques,¹ & Ricardo Vieira^{1,}*

¹Combustion and Propulsion Laboratory, National Institute for Space Research, Cachoeira Paulista, SP, 12630-000, Brazil

²Department of Engineering, Federal University of Lavras (UFLA), Lavras, MG, 37200-000, Brazil

*Address all correspondence to: Ricardo Vieira, Combustion and Propulsion Laboratory, National Institute for Space Research, Cachoeira Paulista, SP, 12630-000, Brazil; Tel.: +55 (12) 31869484, E-mail: ricardo.vieira@inpe.br

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In the last decades there has been an increasing interest in low-cost, low environmental impact liquid propulsion systems, while safety and a short development cycle continue to be essential requirements. This work presents the development of an ethanol and monoethanolamine based rocket fuel, hypergolic with hydrogen peroxide (H_2O_2), featuring low toxicity and high density-specific-impulse. Initially, a set of tests was carried out to evaluate this parameter in a range of compositions prepared with several catalysts dissolved in monoethanolamine. Among the catalytic materials tested, copper nitrate revealed the shortest ignition delay. Subsequently, a Design of Experiments approach was applied to select the ethanol and catalyst content in the fuel capable of decreasing the ignition delay of the reaction. The results showed that an optimized fuel composition with 61% monoethanolamine, 30% ethanol, and 9% hydrated copper nitrate weight percent, yielded a minimum ignition delay of 15.7 ms.

KEY WORDS: *ignition delay, liquid propulsion, hypergolic, green propellants, monoethanolamine, ethanol*

1. INTRODUCTION

Liquid propulsion based on hydrazine and its derivatives as fuel and on nitrogen tetroxide as oxidizer has been largely used in the last decades for space propulsion applications. Hydrazine is an efficient propellant, but its high cost and toxicity are increasingly important issues that affect any trade-off analysis in this highly competitive field (Kang et al., 2017; Zhang et al., 2018). Therefore, as environmental concerns rise, the search for high-performance,

low-toxicity, low-cost alternatives to these highly toxic and expensive propellants are taking place.

Hydrogen peroxide (H_2O_2) has been employed in many space propulsion systems due to its nontoxicity, storability, and safety when compared to other commonly used monopropellants (Shrivignesh et al., 2016; Wu et al., 2015). Although H_2O_2 can also be used in bipropellant rockets, it requires finding a suitable pair that results in hypergolic reaction, low ignition delay, low toxicity, and low cost (Batonneau et al., 2014).

The main advantage of a hypergolic propellant pair is that it allows multiple starts of the rocket engine without complex ignition mechanisms. A low ignition delay is also highly desirable in a hypergolic pair and depends on complex mass and heat transfer (physical factors) and on the kinetics of the combustion reaction (chemical factors) (Alfano et al., 2006). Long ignition delays can produce high pressure peaks that can cause catastrophic failure. Therefore, hypergolic pairs with the lowest possible ignition delay are desired (Davis and Yilmaz, 2014).

The U.S. Naval Air Warfare Center developed a nontoxic fuel called *Block 0* based on a colloidal suspension of manganese oxide in methanol. The *Block 0* is fully miscible and hypergolic with H_2O_2 90 wt.% (Purcell et al., 2002). Later, a research group from Purdue University (Pourpoint and Anderson, 2007) studied the effect of pressure and concentration of H_2O_2 on ignition delay with *Block 0* fuel. The authors observed that the reduction of both the atmospheric pressure and H_2O_2 concentration increases significantly the ignition delay of this reaction. It has also been observed that increasing the catalyst content in the fuel reduces the ignition delay. However, the catalyst content employed in *Block 0* was very high (10–30 wt.%), making it unfeasible in certain applications, either by cost, depending on the amount needed to reduce the ignition delay, or by occasional incrustations formed in the nozzle's throat, generated by the multiphase flow from the combustion of the propellant charged with an excess of manganese oxide.

Despite of its well-known harmful potential, hydrazine has found widespread use in the several last decades both in liquid bipropellant systems and as a monopropellant. Sutton and Biblarz (2016) pointed out that hydrazine has a recommended 8 h personnel exposure limit of 0.01 ppm and that it is known as an animal carcinogen and a suspected carcinogen for people. On the contrary, monoethanolamine has been recognized by Ak et al. (2011) as a low-toxicity fuel. Melof and Grubelich (2001) tested several fuels as hypergolic pairs for H_2O_2 90%, and the best results were obtained with monoethanolamine catalyzed with CuCl_2 . Ak et al. (2011) also studied the hypergolicity of monoethanolamine with H_2O_2 85% with a content of 10% CuCl_2 as catalyst in monoethanolamine (4.7 wt.% of Cu^{2+}), in which case the ignition delay was around 30 ms.

Ethanol by itself does not react spontaneously with H_2O_2 90%, but mixing it with monoethanolamine and an appropriate catalyst makes the reaction hypergolic, thus decreasing the ignition delay. This is due to the combined effect of the higher volatility of ethanol and the decomposition of the H_2O_2 promoted by the formation of a complex between the catalyst and monoethanolamine. Also, adding ethanol reduces the fuel viscosity making it easier to design the propellant injection system. The motivation of this work was to develop a low-toxicity, environmentally benign fuel blend. Hence, this study had two objectives. The first one was to select a salt capable of complexation with monoethanolamine to promote the catalytic decomposition of hydrogen peroxide. The second objective was to apply central composite design to assess the effects of two factors, namely, the composition of the blend ethanol-monoethanolamine and the salt content in the fuel on the ignition delay of the hypergolic reaction.

2. METHODOLOGY

2.1 Concentration of H₂O₂

Starting with a stabilized H₂O₂ 70% (Interlox 70-10) supplied by Solvay Chemicals Inc., an increase in the concentration up to H₂O₂ 90 wt.% was achieved with a counterflow of hot dry air. The concentrator comprised a Jacob's ladder tube heated in a water bath at 370 K. In one side of the tube, hot dry air was injected and on the other side H₂O₂ was introduced. In the U-shaped section of the tube, concentrated H₂O₂ was collected. Hydrogen peroxide was diluted and titrated in acidic medium with a standardized potassium permanganate solution. The equipment was able to produce up to 15 kg of H₂O₂ 90% in 24 h.

2.2 Catalyst Selection

According to Schumb et al. (1955), a large number of inorganic compounds decomposes hydrogen peroxide. In order to choose the appropriate salt to act as the catalyst, a number of salts were added to monoethanolamine, generating a set of distinct fuels, and their hypergolicity was assessed by means of a drop test with H₂O₂ 90%. Initially, the amount of salt added was calculated so that a 2% cation concentration (weight/volume) was kept throughout the experiments. The salts tested are shown in Table 1.

2.3 Drop Test

The drop test was used to check the hypergolicity of the salt-added monoethanolamine fuels listed in Table 1 with H₂O₂ 90%. This test consisted of impinging a single drop of H₂O₂ 90% over a small amount (1 mL) of each of the fuels. The experiments were carried out at 298 K, and the drop height was kept in 25 cm. A high-speed camera (FASTEC model S3) was used to record 3×10^3 fps and measure the induction time of the reaction. The fuel that yielded the shortest ignition time was selected for the subsequent study of adding ethanol to further reduce the propellant ignition delay.

TABLE 1: Catalyst salts used for each experiment

Experiment	Catalyst	Experiment	Catalyst
1	Copper II chloride	11	Nickel sulphate
2	Copper II nitrate	12	Chromium nitrate
3	Manganese nitrate	13	Silver nitrate
4	Manganese acetate	14	Aluminium nitrate
5	Manganese sulphate	15	Iron nitrate
6	Cobalt acetate	16	Lead nitrate
7	Cobalt nitrate	17	Potassium acetate
8	Nickel acetylacetonate	18	Potassium permanganate
9	Nickel chloride	19	Magnesium nitrate
10	Nickel nitrate	20	Sodium bi-sulphite

2.4 Adding Ethanol to Reduce Ignition Delay

Central composite design, one of the most useful tools in the Design of Experiments (DoE) set of techniques, was used to optimize the ethanol load and catalyst content to be added to the fuel. This design was constructed to allow for the fitting of a second-order model, based on the experimental results for the ignition delay. Therefore, the independent variables (factors) considered in this study were the ethanol load (vol./vol.%) (ε_1 , with X_1 coded levels) and the catalyst content in the fuel (wt./vol.%) (ε_2 , with X_2 coded levels). The relationship between the real and coded values is given by

$$X = \frac{\varepsilon - \varepsilon_0}{\Delta} \quad (1)$$

where ε_0 is the real value in the central point of the experimental range, namely 2Δ , which is comprised between the maximum and minimum experimental values initially selected. In the coded values, these limits are represented respectively by 1 and -1 . This initial experimental region is further expanded through a 45 deg rotation, so that the final range of investigation spans from $\sqrt{2}$ to $-\sqrt{2}$, for both variables. Hence, each variable operates at five levels (i.e., $-\sqrt{2}$, -1 , 0 , 1 , $\sqrt{2}$). The real and coded levels of the independent variables are given in Table 2.

The ignition delay (measured in milliseconds) was taken as the response variable. Eleven experiments were carried out, including three replicates of the center point to estimate the pure error needed for the analysis of variance (ANOVA). Once the experiments were performed, the results were fitted to a second-order model to correlate them to the independent variables. The regression coefficients of individual linear, quadratic, and interaction model terms were determined. The appropriateness of the regression was assessed through ANOVA. The resulting model ultimately allowed the construction of a response surface and contour maps, which clearly pointed out the coordinates of the independent variables associated with a region of minimum ignition delay.

3. RESULTS

The ignition delay is related to the reactivity and miscibility between H_2O_2 and the fuel used. From all of the catalysts presented in Table 1, only the copper salts were efficient in the H_2O_2 decomposition, generating a hypergolic reaction. The salts of experiments 3 to 9 produced vigorous but not hypergolic decomposition. The other salts resulted in a slower reaction rate.

In the drop test, once the H_2O_2 reaches the copper complexes in the monoethanolamine, its decomposition generates a considerable amount of heat, promoting the fuel vaporization and leading to the ignition of the system (Lee et al., 2018; Shrivignesh et al., 2016). In this case, the decomposition reaction is catalyzed by the copper complexes formed with monoethanolamine.

TABLE 2: Real and coded levels of independent variables

Symbols	Real Levels					
ε_1	Ethanol load (vol./vol.%)	25.86	30	40	50	54.14
ε_2	Catalyst content (wt./vol.%)	0.94	1.25	2.00	2.75	3.06
—	Coded Levels					
X_1	Ethanol load	$-\sqrt{2}$	-1	0	1	$\sqrt{2}$
X_2	Catalyst content	$-\sqrt{2}$	-1	0	1	$\sqrt{2}$

Copper has many chemical and spectroscopic properties due to its many types of bonding and coordination numbers. Thus, different types of geometries of the cupric salts complexes with monoethanolamine are possible (Casassas et al., 1989).

According to the literature, monoethanolamine (Mea) can complex with copper in many stoichiometric relations to form compounds, such as [Cu(Mea)]²⁺, [Cu(Mea)₂]²⁺, [Cu(Mea)₃]²⁺, and [Cu(Mea)₄]²⁺ (Hoffmann et al., 2003; Pankras et al., 2012). Because of its bifunctionality, monoethanolamine can form coordinated bonding with copper through the oxygen atom of the hydroxyl group and the nitrogen atom of the amine group. However, neutral oxygen is a weak electron-donor and monoethanolamine tends to act as a monodentate ligand when the OH group is protonated (i.e., despite being a bifunctional compound) in most reactions monoethanolamine shows amine and ammonia characteristics (Pankras et al., 2012).

The H₂O₂ decomposition begins over the Cu(II)-monoethanolamine complex producing O₂, steam, and heat. The heat generated causes the fuel volatilization and leads to the self-ignition of the propellants in order for their combustion to take place (Bonifacio et al., 2011; Lee et al., 2018). Self-ignition may occur according to three mechanisms: chain branching, thermal, or a combination of both. The ignition by chain branching is based on initiation, branching, propagation, and chain termination reactions, where the reaction rate increases due to the branching steps, which cause the exponential growth of the chain propagation centers. The thermal mechanism is based on the concept that, when the rate of release of thermal energy by the reaction is greater than the rate of dissipation of thermal energy (loss), there is an explosive or ignition condition. When the opposite occurs, the thermal explosion is impossible. When the two rates are equal, the critical condition of ignition is attained (Glassman and Yetter, 2008). Fuels catalyzed with copper salts were the only ones that allowed the system to auto ignite, as revealed by the drop test. Between the copper salts assessed, the nitrate featured the lowest ignition delay. Therefore, it was chosen for the continuation of the work.

3.1 Ethanol Addition: Fuel Optimization

The optimization approach adopted in this study allows the building of sequential experiments, in order to quantify the influence of the selected factors on the response variable. Two factors at five levels each were considered as independent variables in the study. The central point in real values consists of 40 vol./vol.% of ethanol in the fuel and 2 wt./vol.% of Cu²⁺ in the fuel. Table 3 shows the 11 experimental results that were used to evaluate the response variables (ignition delay).

The data obtained were fitted to a quadratic model as described by Bezerra et al. (2008). It is described in Eq. (2) in terms of coded factors (X_1 and X_2), where τ is the response variable (i.e., the ignition delay).

$$\tau = 20.445 + 2.114X_1 - 7.467X_2 + 2.638X_1^2 + 6.955X_2^2 - 2.266X_1X_2 \quad (2)$$

Table 4 presents the ANOVA for the fitted quadratic model expressed by Eq. (2). The significance of the regression can be evaluated by the ratio between the values of the quadratic regression (MS_{Adj}) and the quadratic residue (MS_{Res}), F_{calc} . The comparison of F_{calc} to the critical F-value (F_{crit}) tabulated for the 95% confidence level shows that $F_{calc} > F_{crit}$. Hence, the fitted model described by Eq. (2) is significant. An important way to confirm the model adequacy is the lack of fit F-test. Here, the idea is to calculate the ratio between the values of the quadratic lack of fit (MS_{lof}) and the quadratic error (MS_{err}), both of them unfolding from the quadratic residue. As before, the resulting ratio, F_{lof} , is then compared to the critical F-value

TABLE 3: Central composite design-coded dependent variables, corresponding experimental conditions and response

Exp.	X_1	X_2	ε_1 (% Ethanol)	ε_2 (% Cu^{2+})	τ (ms)
1	-1	-1	30.00	1.25	33.26
2	-1	1	30.00	2.75	22.00
3	1	-1	50.00	1.25	42.00
4	1	1	50.00	2.75	21.66
5	0	0	40.00	2.00	19.66
6	0	0	40.00	2.00	20.66
7	0	0	40.00	2.00	21.00
8	$-\sqrt{2}$	0	25.86	2.00	23.00
9	0	$\sqrt{2}$	40.00	3.06	24.66
10	$\sqrt{2}$	0	54.14	2.00	29.00
11	0	$-\sqrt{2}$	40.00	0.94	44.50

TABLE 4: Analysis of variance (ANOVA)

Source	DF	SS_{Adj}	MS_{Adj}	F_{calc}	F_{crit}
Regression	5	714.14	154.83	240.42	5,05
X_1	1	35.63	35.63	55.33	6.61
X_2	1	444.73	444.73	690.60	6.61
X_1^2	1	38.97	38.97	60.52	6.61
X_2^2	1	270.94	270.94	420.72	6.61
$X_1 \cdot X_2$	1	20.55	20.55	31.91	6.61
Residue	5	3.22	0.64	—	—
Lack of fit	3	2.26	0.75	1.56	9.55
Error	2	0.96	0.48	—	—
Total	10	777.36	—	—	—

(F_{crit}) tabulated for the 95% confidence level, but in this case the adequacy of the fit is expressed by $F_{\text{lof}} < F_{\text{crit}}$. Accordingly, as presented in Table 4, $F_{\text{lof}} = 1.56 < F_{\text{crit}} = 9.55$ meaning that the model does not present lack of fit.

At this point, it might be instructive to perform a qualitative assessment of the fitted model. In doing so, it is important to consider that the sign of a coefficient ($\pm$) in the fitted model defines as to whether or not its contribution to the response is positive or negative as one goes through the experimental range from the -1 to 1 coded level. It is worthwhile noting that the sign of the interaction term $X_1 \cdot X_2$ is negative. Hence, the factors have an antagonistic interaction effect on the response variable. Thus, regarding only their interaction term, their simultaneous increase will reduce the ignition delay. Moreover, as the X_2 linear term bears a negative sign, its increase will also contribute to reduce the ignition delay. On the other hand, the X_1 factor presents an opposite behavior (i.e., as one goes from its -1 to 1 coded level, the response variable increases). These

remarks suggest that the attainment of a minimum ignition delay region implies the adjustment of the factors somewhere next to the central point.

Figure 1 presents a graphical display of the model, respectively, as contour plots and response surface plot, providing a clear picture of the relationship between the response and the experimental levels for each variable. It is possible to calculate the coordinates of the point of minimum through the system formed by the partial first derivatives of the fitted model [Eq. (2)]. When both these equations are equaled to zero, the system is complete. Solving it leads to the aforementioned coordinates.

Therefore, the values calculated for the coded variables X_1 and X_2 , were -0.183 and 0.507 , respectively. Using Eq. (1), the values of the natural variables, ε_1 and ε_2 , were 38.17 and 2.38% , respectively. When replacing the coded optimal values of X_1 and X_2 in Eq. (2), the model predicts a minimum ignition delay of 18.3 ms. This result agrees with the qualitative assessment discussed previously. In order to determine the mass composition of the fuel, the monoethanolamine and ethanol volumes were converted to mass contents by their respective densities. In addition, the mass of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ was calculated from the Cu^{2+} content (2.38 wt.%) predicted by the model. From the mass of each component of the mixture, it was possible to define that the fuel should contain, by mass, $\sim 61.0\%$ of monoethanolamine, 30.1% of ethanol and 8.9% of $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$.

Finally, a drop test reacting H_2O_2 90% and the optimized fuel resulted in an ignition delay on the order of 15.7 ms (Fig. 2). It is important to note that drop tests were also conducted with H_2O_2 90% and monoethanolamine containing only 8.9% copper nitrate. In this test, the ignition delay result was 36 ms. The addition of ethanol to the fuel reduced the ignition delay in the hypergolic reaction due to its lower flash point as compared to monoethanolamine.

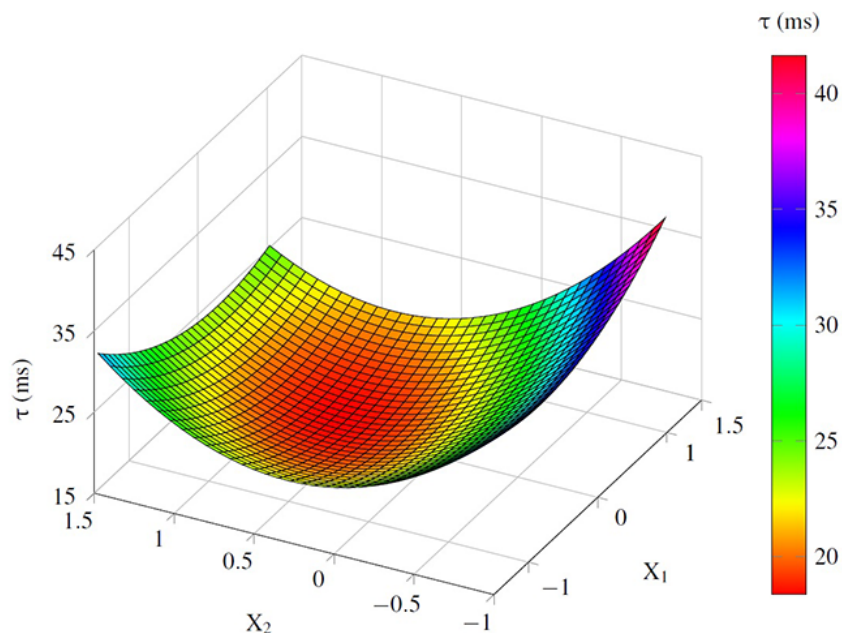


FIG. 1: Response surface plot showing the ignition delay as a function of the amounts of ethanol and Cu^{2+}

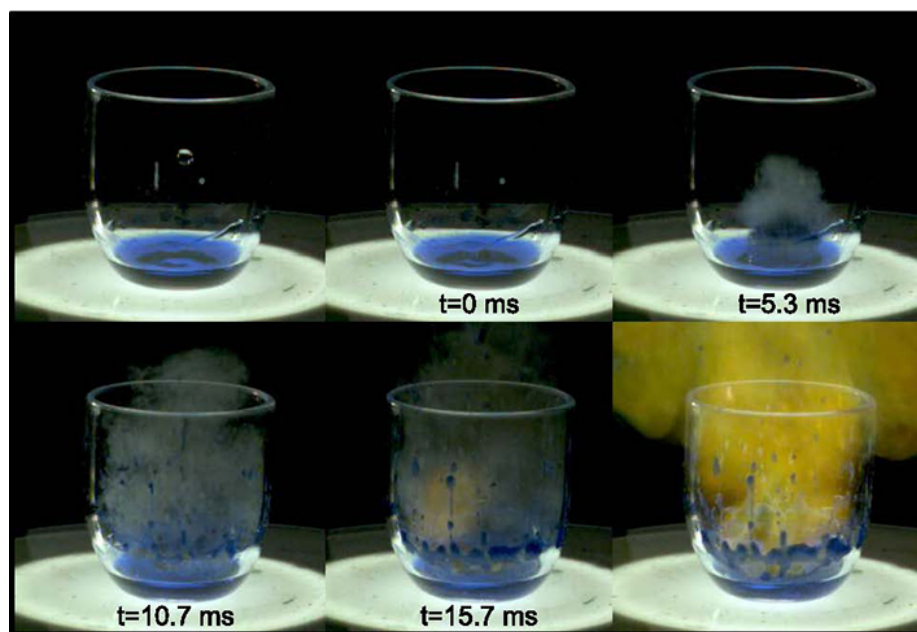


FIG. 2: Drop test of the optimized fuel with H_2O_2 90%

4. CONCLUSIONS

Different catalytic materials were added to monoethanolamine, but only the copper salts were able to promote an effective decomposition of H_2O_2 90%. Copper nitrate yielded the least ignition delay in this reaction. Central Composite Design has been successfully applied to the optimization of the ethanol load and catalyst content in monoethanolamine to attain the fastest hypergolic reaction with H_2O_2 90%. The F test, at 95% confidence level, showed that the fitted quadratic model was significant. The results suggested that the composition of the fuel with the lowest ignition delay (15.7 ms) should contain about 61.0% monoethanolamine, 30.1% ethanol and 8.9% $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$ by weight. Besides being cost effective, results suggested that the addition of ethanol also causes a significant decrease in the ignition delay, due to a reduction in the flash point of the fuel. Furthermore, the green aspect of the fuel blend developed in this study was attained not only by ethanol itself, but by monoethanolamine as well, as long as its toxicity is lower than that of hydrazine.

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REFERENCES

- Ak, M., Ulas, A., Sümer, B., Yazıcı, B., Yıldırım, C., Gönc, L., and Orhan, F., (2011) An Experimental Study on the Hypergolic Ignition of Hydrogen Peroxide and Ethanolamine, *Fuel*, **90**(1), pp. 395–398.

- Alfano, A.J., Mills, J.D., and Vaghjiani, G.L., (2006) Highly Accurate Ignition Delay Apparatus for Hypergolic Fuel Research, *Rev. Sci. Instrum.*, **77**(045109), pp. 1–5.
- Batonneau, Y., Brahmi, R., Cartoixa, B., Farhat, K., Kappenstein, C., Keav, S., Kharchafi-Farhat, G., Pirault-Roy, L., Saouabe, M., and Scharlemann, C., (2014) Green Propulsion: Catalysts for the European FP7 Project GRASP, *Top. Catalys.*, **57**(6-9), pp. 656–667.
- Bezerra, M.A., Santelli, R.E., Oliveira, E.P., Villar, L.S., and Escalera, L.A., (2008) Response Surface Methodology (RSM) as a Tool for Optimization in Analytical Chemistry, *Talanta*, **76**(5), pp. 965–977.
- Bonifacio, S., Festa, G., and Sorge, A.R., (2011) Experimental Assessment of Hydrogen Peroxide Decomposition in a Monopropellant Thruster, *Int. J. Energetic Mater. Chem. Propuls.*, **10**(6), pp. 497–522.
- Casassas, E., Gustems, L.L., and Tauler, R., (1989) Spectrophotometric Study of Complex Formation in Copper(II) Mono, Di, and Triethanolamine Systems, *J. Chem. Soc. Dalton Trans.*, pp. 569–573.
- Davis, S.M. and Yilmaz, N., (2014) Advances in Hypergolic Propellants: Ignition, Hydrazine, and Hydrogen Peroxide Research, *Adv. Aerospace Eng.*, **2014**(1), pp. 1–9.
- Glassman, I. and Yetter, R.A., (2008) *Principles of Combustion*, 4th ed., Burlington: Academic Press.
- Hoffmann, S.K., Goslar, J., Polus, I., and Mazela, B., (2003) Spectroscopic Studies of Electronic Structure of Copper(II) Ethanolamine Complexes in Solutions and in Impregnated Wood, *Appl. Mag. Reson.*, **24**(3), pp. 321–331.
- Kang, H., Won, J., Baek, S.W., and Kwon, S., (2017) Autoignition and Combustion Characteristics of Sodium Borohydride-based Non-Toxic Hypergolic Fuel Droplet at Elevated Temperatures, *Combust. Flame*, **181**(1), pp. 149–156.
- Lee, K., Han, G., Cho, S., and Bae, J., (2018) Pressurized Diesel Fuel Processing using Hydrogen Peroxide for the Fuel Cell Power Unit in Low-Oxygen Environments, *J. Power Sources*, **380**(1), pp. 37–45.
- Melof, B. and Grubelich, M., (2001) Investigation of Hypergolic Fuels with Hydrogen Peroxide, in *37th Joint Propulsion Conference and Exhibit*, American Institute of Aeronautics and Astronautics (AIAA), pp. 1–10.
- Pankras, S., Cooper, P.A., and Wylie, S., (2012) Relationship between Copper Species in Solution and Leaching from Alkaline Copper Quat (ACQ) Treated Wood, *Holzforschung*, **66**(4), pp. 505–514.
- Pourpoint, T.L. and Anderson, W.E., (2007) Hypergolic Reaction Mechanisms of Catalytically Promoted Fuels with Rocket Grade Hydrogen Peroxide, *Combust. Sci. Technol.*, **179**(10), pp. 2107–2133.
- Purcell, N., Diede, A., and Minthorn, M., (2002) Test Results of New Reduced-Toxicity Hypergols for Use with Hydrogen Peroxide Oxidizers, in *5th Int. Hydrogen Peroxide Propulsion Conf.*, Universidade de Purdue, pp. 111–122.
- Schumb, W.C., Satterfield, C.N., and Wentworth, R.L., (1955) *Hydrogen Peroxide*, New York: Reinhold Publishing Corporation.
- Shrivignesh, K., Suwathy, R., Thirumalaikumar, S.K., and Venkatesan, M., (2016) Annular Mixing Section Effects on Hydrogen Peroxide-Air Two-Phase Flow inside a Mini-Channel, *Int. J. Energetic Mater. Chem. Propuls.*, **15**(6), pp. 453–466.
- Sutton, G.P. and Biblarz, O., (2016) *Rocket Propulsion Elements*, 9th ed., Hoboken, NJ: Wiley.
- Wu, C., Wang, X., Zhou, X., Yang, T., and Zhang, T., (2015) Supported MnO_x/SrO-Al₂O₃ High-Cell Density Honeycomb Ceramic Monolith Catalyst for High-Concentration Hydrogen Peroxide Decomposition, *Int. J. Energetic Mater. Chem. Propuls.*, **14**(5), pp. 421–436.
- Zhang, X., Pan, L., Wang, L., and Zou, J.J., (2018) Review on Synthesis and Properties of High-Energy-Density Liquid Fuels: Hydrocarbons, Nanofluids and Energetic Ionic Liquids, *Chem. Eng. Sci.*, **180**(1), pp. 95–125.